Detecting Delay Cascades in the U.S. Airline Network

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GitHub Repository **Project Video**

1 Introduction

Flight delay propagation is one of the number one issues in the commercial airline industry. Each year, the economy loses $33 billion, according to a study done by UC Berkeley, with about half of that burden being because passengers lost time. A single aircraft typically files four or five times a day, carrying the same crew between cities. When one flight is late, the aircraft will arrive late at its next station. As a result, the crew may run out of legal duty time, and the disturbances spread through to other flights further down the stream.

2 Research Question

The goal of this paper is to answer the following three questions:

Which airports and routes are most influential to delays?

How do delays propagate through the network?

What combination of conditions reliably predicts large-scale cascades?

3 Methods – data description

The dataset used for this study was taken from the U.S. Bureau of Transportation Statistics. There are a little over 500,000 domestic flight records per month. Each recorded in the data set represents a single flight, information about the airline, origin, destination, scheduled and actual departure times, as well as delay variables that expalin the reason for the delay. Some of the key variables that are included in my research of Temperature delay (DepDelay) and Arrival delay (ArrDelay) are both measured in minutes. Delay causes included weather, carrier, airspace congestion, and late aircraft delays. Flight identifiers such as tail number, origin, and destination airports were also present in this data set.

When it came to analyzing the behavior of the overall system, what was done was that the data was combined into daily features, where each row represented a single day instead of each row representing a single flight. These features allowed for the capture of the overall system condition for the day. This also helped with network delay propagation and detecting large-scale cascading events. An example of an event of this magnitude might be a thunderstorm that hits one of the major hubs for hours, causing hundreds of delays or canceled flights. This was much more effective and efficient than focusing on individual flights because there are 25,000 - 30,000 us domestic flights daily.

4 Methods – algorithms, implementations, and software

To answer the questions above and analyze delay propagation and identify cascading conditions, many different data science techniques were used. First off, the airport system was modeled as a directed weighted network/graph, where the airports represented the nodes and the flights represented the edges. Out of the 12 million rows, there were 356 different airports and 7257 different routes connecting these nodes. PageRank algorithm was the first applied to this network to identify the influence of each airport in terms of how it spreads delays; this helped later on to identify what airports really amplify delays within the system as well as what one supress delays. A second PageRank model weighed routes by the number of flights in order to measure traffic importance. Comparing these two PageRank scores helps identify what airports amplify delays beyond what their traffic volume would generally predict.

To dive a little deeper, a carrier hub analysis was also completed. For each airport, the share of flights operated by the combined carrier was calculated. It was intriguing to try and understand if there was a correlation between hub and airline share. This proved insightful between single and multi-carrier hubs and evaluated whether carrier concentration influenced delay propagation. This helps explain why some high-traffic airports absorb delays effectively while others amplify them.

Next, Apriori association rule mining was used to study cascade conditions. Flight records were pulled together into daily systems and then converted into a binary signal, for example, high weather delay, high late aircraft delay, high NAS delay, and high carrier delay. It was then used to determine cascade days that were defined as the top 10 percent of data by total delay. Apriori was then used to identify combinations of conditions that occur with cascading days reliably. This was then evaluated using support, confidence, and lift. All analysis for this was done in Python using pandas and NumPy for data processing, NetworkX for graph and PageRank, mlxtend for association rule mining, and matplotlib for visualization.

5 Results

Using a delay-weighted page rank model, it was identified that the most influential airports in spreading delays were airports like DFW, ORD, and DEN. These are the most influential airports in propagating delays across the network, with DFW standing out as the outlier. Delay influence was compared to traffic importance and revealed a clear distinction between airports that amplified delays and those that absorbed them. Airports such as DFW, ORD, and CLT created more delays than their traffic volume would have predicted, while airports like ATL and LAX absorbed delays more effectively. This was identified by comparing the inbound traffic at airports where the delay rank is higher then traffic, meaning it is pulling more delay then it's traffic would suggest.

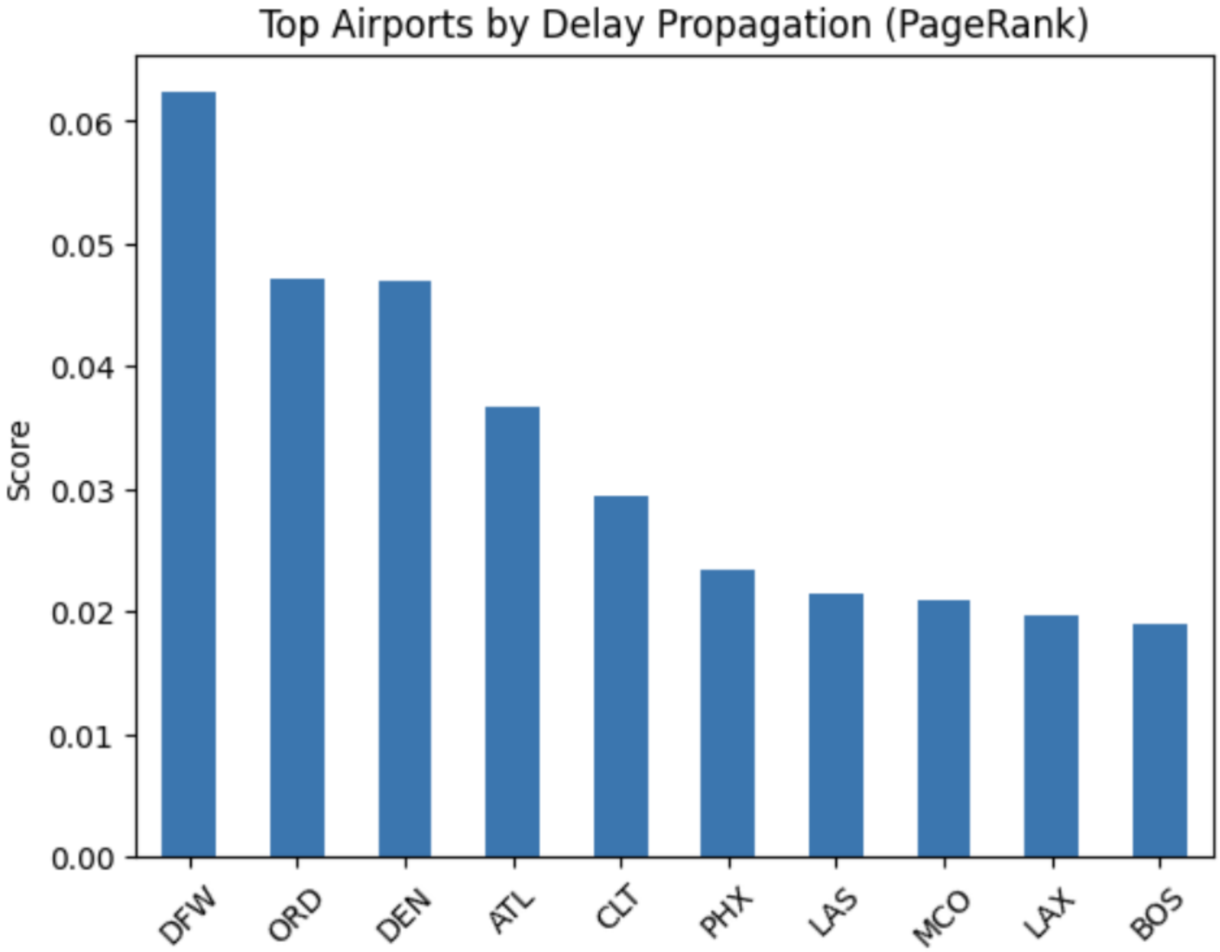

Figure 1: Top Airports by Delay Propagation Influence (PageRank Scores)

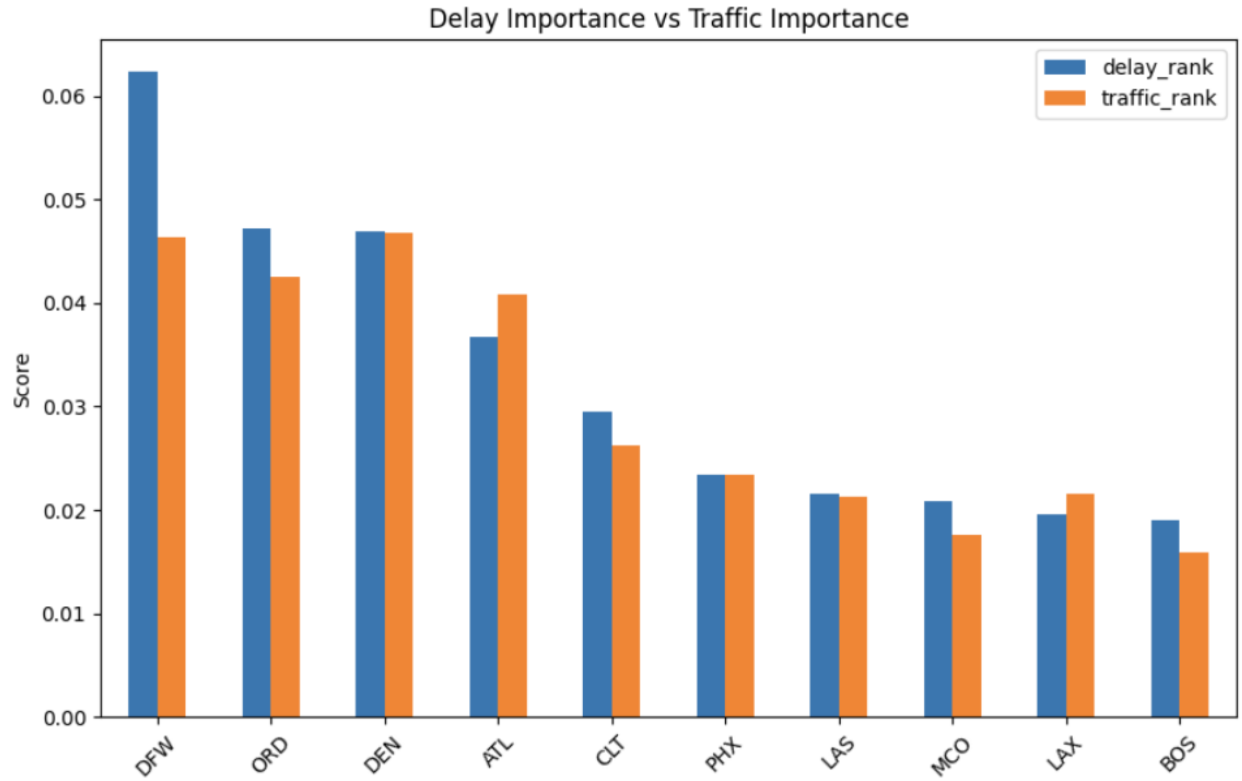

Figure 2: Comparison of Delay Propagation and Traffic Importance Across Major U.S. Airports

To try to explain these differences, carrier concentration at major hubs was looked at. Airports with dominant carriers like ATL with Delta tend to operate more efficiently and absorb delays. In contrast, airports with more split carrier presence show greater delay amplification. But it was determined that the relationship is not absolute, because the variance within carriers indicated that there are other operational factors that also contribute to delay propagation. It was discovered that delay propagation is primarily driven by the rotation of aircraft. Average departure delays increase steadily throughout the day, with evening delays often reaching four to five times higher than morning delays. This can be explained by late aircraft delays; as a result, this becomes the dominant delay category by the afternoon and accounts for the largest share of delays in the network.

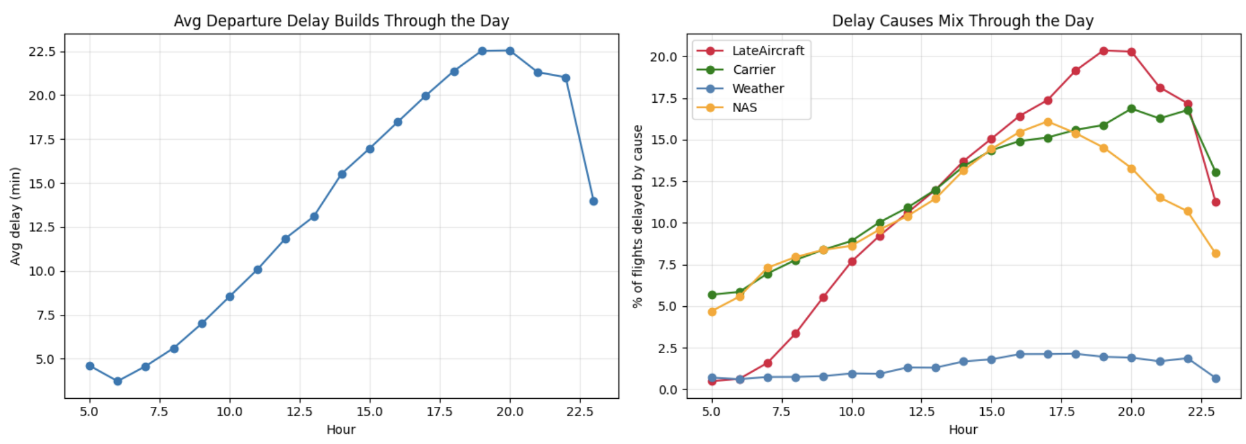

Figure 3: Growth of Departure Delays and Delay Causes Over the Course of the Day

Apriori association rule mining was used to identify different combinations of conditions that reliably predict cascade days. The strictest/strongest rules achieved 65% lift and a lift of 6.5; this means that there is a strong relationship between system conditions and large scale disruptions. A key finding was that no high confidence cascade predicting rule was found without weather as a component. Operational factors like late aircraft, NAS, and carrier delayed also contribute to cascades; they do not reliably predict them on their own. Weather acts as a necessary trigger, while operational stress increases its effects.

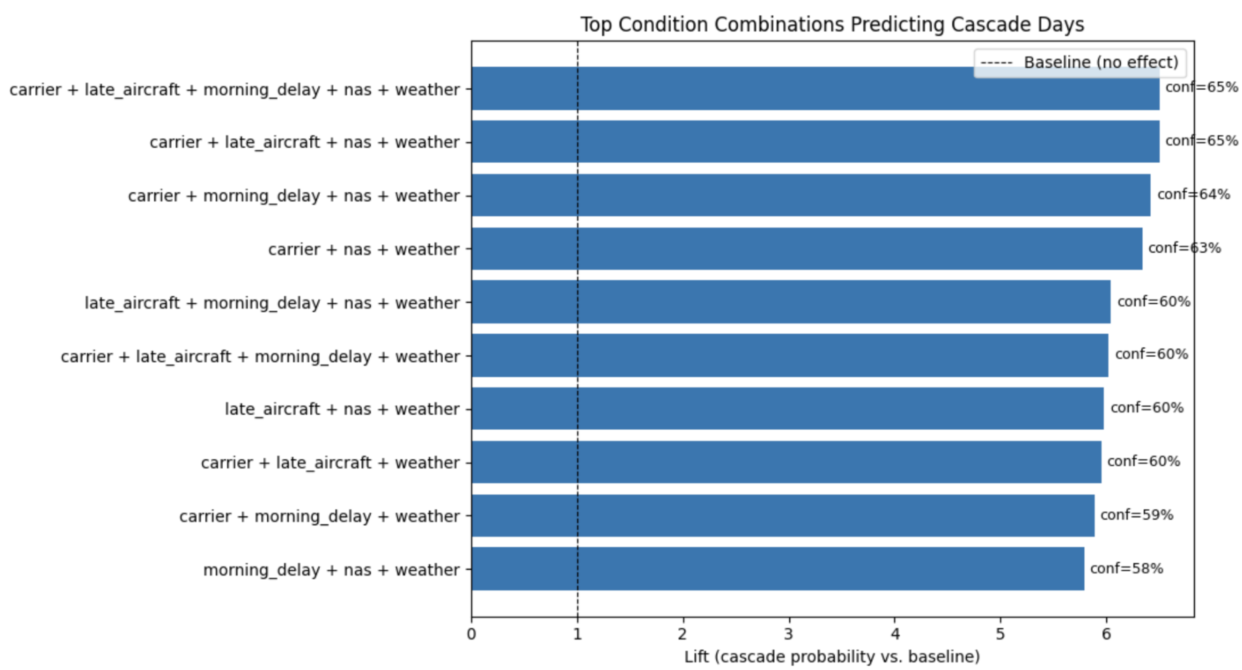

Figure 4: Top Condition Combinations Predicting Cascade Days (Apriori Analysis)

6 Limitations and future work

This study does a good job of providing information on how delay propagation across the U.S. airline network, but there are still several limitations that are worth being noted.

First off, the data set relies on self-reported delay causes, which may introduce insensitivity and bias. Airlines categorize delays into carrier, late aircraft, weather, and NAS, but these categories can overlap and are not always assigned consistently. As a result, the same delay causes are often misrepresented.

Second, this analysis is based on observation data, which limits the ability to make causal claims. While methods such as PageRank and Apriori identify a strong amount of association and patterns, they are not reliable for providing direct causes.

Finally, the predictive modeling component was only trained on a little over a year's worth of data. This may limit the ability to generalize across many years or unusual disruptions. There are many ways to extend this work.

Incorporating external weather data, it would allow for a more precise understanding of how weather events trigger cascades, rather than relying on reported delay categories.

Expanding the predictive model to include longer time periods and seasonal patterns, it would likely improve the practical applicability.

Finally, it may be worth diving into a deeper analysis of aircraft operations schedules and crew contents, as this would provide more detailed exploration of the mechanistic nature behind delay prediction and amplification.

Citations

References

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